

Homework 05

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Question 1

Given a relation schema $Q(ABCDGH)$ and a set of FDs F is specified on Q as follows,

$F = \{ f1: GH \rightarrow A, \\ f2: AG \rightarrow B, \\ f3: CD \rightarrow G, \\ f4: HG \rightarrow D, \\ f5: BH \rightarrow C, \\ f6: CD \rightarrow H, \\ f7: C \rightarrow A \}$

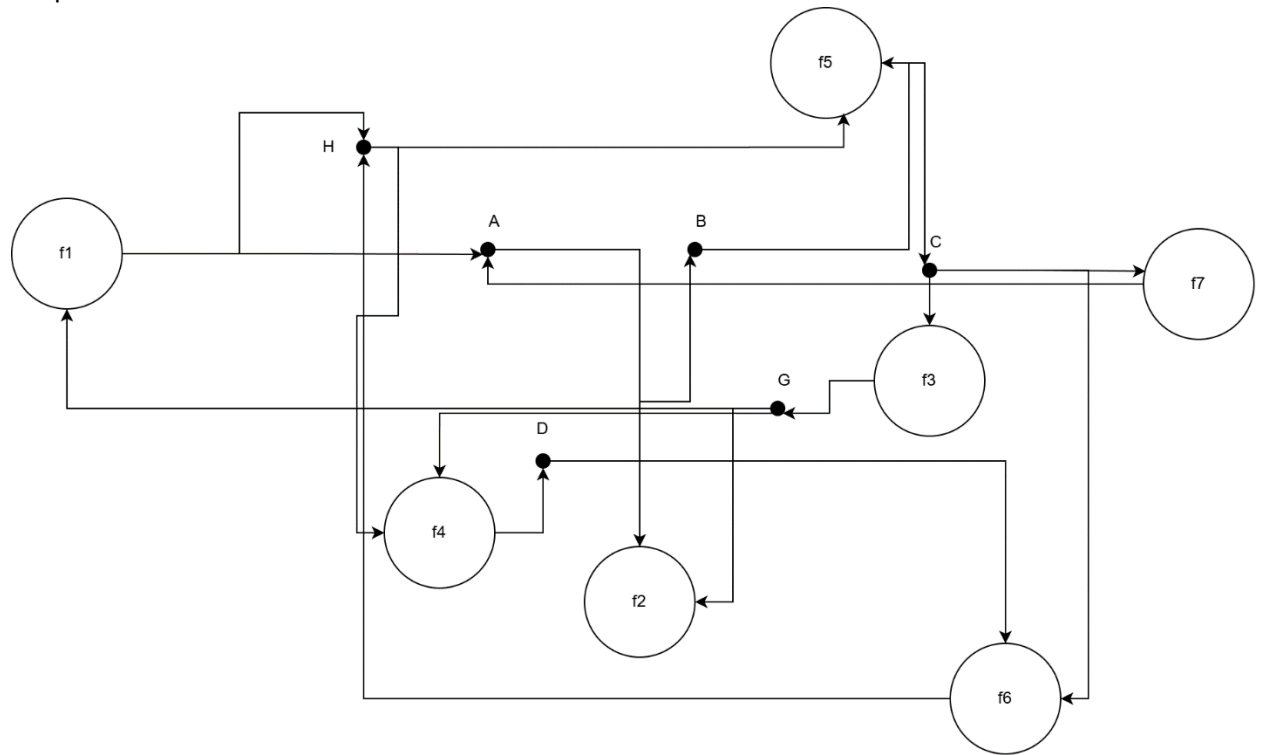
Evaluate the normal form of Q . Support your answer.

Consider F is good or not?

- Split the right-hand sides attributes $\rightarrow F$ is already satisfied
- Remove extraneous attributes in the left-hand sides
 - $GH \rightarrow A$: $\{G\}^+ = \{G\}$ does not contain H , and $\{H\}^+ = \{H\}$ does not contain G , so we keep $GH \rightarrow A$
 - $AG \rightarrow B$: $\{G\}^+ = \{G\}$ does not contain A , and $\{A\}^+ = \{A\}$ does not contain G , so we keep $AG \rightarrow B$
 - $CD \rightarrow G$: $\{C\}^+ = \{C, A\}$ does not contain D , and $\{D\}^+ = \{D\}$ does not contain C , so we keep $CD \rightarrow G$
 - $HG \rightarrow D$: $\{G\}^+ = \{G\}$ does not contain H , and $\{H\}^+ = \{H\}$ does not contain G , so we keep $HG \rightarrow D$
 - $BH \rightarrow C$: $\{B\}^+ = \{B\}$ does not contain H , and $\{H\}^+ = \{H\}$ does not contain B , so we keep $BH \rightarrow C$
 - $CD \rightarrow H$: $\{C\}^+ = \{C, A\}$ does not contain D , and $\{D\}^+ = \{D\}$ does not contain C , so we keep $CD \rightarrow H$
- Remove excessive functional dependencies
 - $\{GH\}^+_{F-\{GH \rightarrow A\}} = \{G, H, D\}$ does not contain A , so we keep $GH \rightarrow A$
 - $\{AG\}^+_{F-\{AG \rightarrow B\}} = \{A, G\}$ does not contain B , so we keep $AG \rightarrow B$
 - $\{CD\}^+_{F-\{CD \rightarrow G\}} = \{C, D, H, A\}$ does not contain G , so we keep $CD \rightarrow G$
 - $\{HG\}^+_{F-\{HG \rightarrow D\}} = \{G, H, A\}$ does not contain D , so we keep $HG \rightarrow D$
 - $\{CD\}^+_{F-\{CD \rightarrow H\}} = \{C, D, G, A\}$ does not contain H , so we keep $CD \rightarrow H$
 - $\{C\}^+_{F-\{C \rightarrow A\}} = \{C\}$ does not contain A , so we keep $C \rightarrow A$

$\rightarrow F$ is minimal.

Graph of FDS:



Since there is no any source in the graph of FDs, meaning that there is no key with only 1 attribute.

From the set of FDs F , we have closure of some set of attributes:

- $\{G, H\}^+ = \{G, H, A, D, B, C\} \rightarrow$ We don't need to consider the combination that include G, H
- $\{A, G\}^+ = \{A, G, B\}$
- $\{A, G, C\}^+ = \{A, G, B, C\}$
- $\{A, G, D\}^+ = \{A, G, B, D\}$
- $\{A, G, C, D\}^+ = \{A, G, B, C, D, H\}$
- $\{C, D\}^+ = \{C, D, G, H, A, B\} \rightarrow$ We don't need to consider the combination that include C, D
- $\{B, H\}^+ = \{B, H, C, A\}$
- $\{B, H, D\}^+ = \{B, H, D, C, A, G\} \rightarrow$ We don't need to consider the combination that include B, H, D

With $\{A, G, C, D\}$ and $\{C, D\}$, we choose $\{C, D\}$ as key because it is minimal. So, the key of relation schema Q is GH, CD, BHD

Evaluate the Normal Form:

- Q does not satisfy the BCNF: there is a functional dependency whose left hand side does not contain the key ($f5: BH \rightarrow C$)

- Q does not satisfy the 3NF: there is a functional dependency whose right-hand side is not a member of the key. (f7: $C \rightarrow A$)
- Q does not satisfy the 2NF: there is a functional dependency whose non-prime attributes are not fully dependent on the key. (f7: $C \rightarrow A$)

So, we conclude that Q satisfies the 1NF.

Question 2

Given a relation schema $Q (ABDELMNOPX)$ and a set of FDs F is specified on Q as follows,

$F = \{f1: AE \rightarrow BDLMNOPX,$
 $f2: LOP \rightarrow AE,$
 $f3: BM \rightarrow N,$
 $f4: OP \rightarrow BD,$
 $f5: NO \rightarrow P\}$

- a) Assume Q has 4 keys, AE , LOP , $BMLO$, and NLO . Let's evaluate the normal form of Q . Support your answer.

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- b) Improve Q to the 3rd normal form by a decomposition $D = \{Q1, Q2\}$, based on the functional dependency $OP \rightarrow D$.

- c) Does D satisfy two properties of lossless join and dependency preservation? Support your answer.

Consider F is good or not:

- Split the right-hand sides attributes

$F' = \{AE \rightarrow B, AE \rightarrow D, AE \rightarrow L, AE \rightarrow M, AE \rightarrow N, AE \rightarrow O, AE \rightarrow P, AE \rightarrow X, LOP \rightarrow A, LOP \rightarrow E, BM \rightarrow N, OP \rightarrow B, OP \rightarrow D, NO \rightarrow P\}$

- Remove extraneous attributes in the left-hand sides
 - $AE \rightarrow B$: $\{A\}^+ = \{A\}$ does not contain E , $\{E\}^+ = \{E\}$ does not contain A , so we keep $AE \rightarrow B$
 - Similarly, we keep $AE \rightarrow D, AE \rightarrow L, AE \rightarrow M, AE \rightarrow N, AE \rightarrow O, AE \rightarrow P, AE \rightarrow X$
 - $LOP \rightarrow A$: $\{OP\}^+ = \{OPBD\}$ does not contain L , $\{LO\}^+ = \{LO\}$ does not contain P , $\{LP\}^+ = \{LP\}$ does not contain O , so we keep $LOP \rightarrow A$
 - $LOP \rightarrow E$: $\{OP\}^+ = \{OPBD\}$ does not contain L , $\{LO\}^+ = \{LO\}$ does not contain P , $\{LP\}^+ = \{LP\}$ does not contain O , so we keep $LOP \rightarrow E$

- $BM \rightarrow N$: $\{B\}^+ = \{B\}$ does not contain M, $\{M\}^+ = \{M\}$ does not contain B, so we keep $BM \rightarrow N$
- $OP \rightarrow B$: $\{O\}^+ = \{O\}$ does not contain P, $\{P\}^+ = \{P\}$ does not contain O, so we keep $OP \rightarrow B$
- $OP \rightarrow D$: $\{O\}^+ = \{O\}$ does not contain P, $\{P\}^+ = \{P\}$ does not contain O, so we keep $OP \rightarrow D$
- $NO \rightarrow P$: $\{O\}^+ = \{O\}$ does not contain N, $\{N\}^+ = \{N\}$ does not contain O, so we keep $NO \rightarrow P$
- Remove excessive functional dependencies
 - $\{AE\}^+_{F' - \{AE \rightarrow B\}} = \{AEBDLMNOPX\}$ contains B, so we remove $AE \rightarrow B$
 - $\{AE\}^+_{F' - \{AE \rightarrow D\}} = \{AEBDLMNOPX\}$ contains D, so we remove $AE \rightarrow D$
 - $\{AE\}^+_{F' - \{AE \rightarrow L\}} = \{AEBDMNOPX\}$ does not contain L, so we keep $AE \rightarrow L$
 - $\{AE\}^+_{F' - \{AE \rightarrow M\}} = \{AEBDLNOPX\}$ does not contain M, so we keep $AE \rightarrow M$
 - $\{AE\}^+_{F' - \{AE \rightarrow N\}} = \{AEBDLMNOPX\}$ contains N, so we remove $AE \rightarrow N$
 - $\{AE\}^+_{F' - \{AE \rightarrow O\}} = \{AELMPX\}$ does not contain O, so we keep $AE \rightarrow O$
 - $\{AE\}^+_{F' - \{AE \rightarrow P\}} = \{AELMOX\}$ does not contain P, so we keep $AE \rightarrow P$
 - $\{AE\}^+_{F' - \{AE \rightarrow X\}} = \{AELMOP\}$ does not contain X, so we keep $AE \rightarrow X$
 - $\{LOP\}^+_{F' - \{LOP \rightarrow A\}} = \{LOPEBD\}$ does not contain A, so we keep $LOP \rightarrow A$
 - $\{LOP\}^+_{F' - \{LOP \rightarrow E\}} = \{LOPABD\}$ does not contain E, so we keep $LOP \rightarrow E$
 - $\{BM\}^+_{F' - \{BM \rightarrow N\}} = \{BM\}$ does not contain N, so we keep $BM \rightarrow N$
 - $\{OP\}^+_{F' - \{OP \rightarrow B\}} = \{OPD\}$ does not contain B, so we keep $OP \rightarrow B$
 - $\{OP\}^+_{F' - \{OP \rightarrow D\}} = \{OPB\}$ does not contain D, so we keep $OP \rightarrow D$
 - $\{NO\}^+_{F' - \{NO \rightarrow P\}} = \{NO\}$ does not contain P, so we keep $NO \rightarrow P$

Then we have:

$F'' = \{AE \rightarrow L, AE \rightarrow M, AE \rightarrow O, AE \rightarrow P, AE \rightarrow X, LOP \rightarrow A, LOP \rightarrow E, BM \rightarrow N, OP \rightarrow B, OP \rightarrow D, NO \rightarrow P\}$

After rewriting we have:

$F'' = \{AE \rightarrow LMOPX, LOP \rightarrow AE, BM \rightarrow N, OP \rightarrow BD, NO \rightarrow P\}$

We see that $F \neq F''$, so F is not minimal

a) Assume Q has 4 keys, AE, LOP, BMLO, and NLO. Let's evaluate the normal form of Q. Support your answer.

- Q does not satisfy the BCNF: there is a functional dependency whose left-hand side does not contain the key ($NO \rightarrow P$)
- Q does not satisfy the 3NF: there is a functional dependency whose right-hand side is not a member of the key ($OP \rightarrow BD$)
- Q does not satisfy the 2NF: there is a functional dependency whose non-prime attributes are not fully dependent on the key ($OP \rightarrow BD$)

So, we conclude that Q satisfies the 1NF

b) Improve Q to the 3rd normal form by a decomposition $D = \{Q1, Q2\}$, based on the functional dependency $OP \rightarrow D$

- The functional dependency $OP \rightarrow D$ is causing Q to violate 3NF because OP is not a key, and D is a non-prime attribute. We will split Q into Q1 and Q2

- Q1 (OPD)
 - Key: OP
 - $F1 = \{OP \rightarrow D\}$
- Q2 (AEBLMNOPX)
 - Key: AE, LOP, BMLO, and NLO
 - $F2 = \{AE \rightarrow LMOPX, LOP \rightarrow AE, BM \rightarrow N, OP \rightarrow B, NO \rightarrow P\}$

Check whether Q1 and Q2 satisfy 2NF or not:

- Q1 satisfies 2NF because with all functional dependencies, their non-prime attributes are fully functionally dependent on the key
- Q2 satisfies 2NF because with all functional dependencies, their non-prime attributes are fully functionally dependent on the key

Check whether Q1 and Q2 satisfy 3NF or not:

- Q1 (OPD)
 - OP is key $\rightarrow$ O, P is prime attribute and D is non-prime attribute
 - $OP \rightarrow D$: OP is key, and D is non-prime attribute $\rightarrow$ satisfy 3NF
- Q2 (ABELMNOPX)
 - AE, BMLO, LOP, NLO are key $\rightarrow$ A, B, E, M, N, L, O, P is prime attribute and X is non-prime attribute
 - $AE \rightarrow LMNOPX$: AE is key, and X is non-prime attribute $\rightarrow$ satisfy 3NF
 - $LOP \rightarrow AE$: AE, LOP are both key $\rightarrow$ satisfy 3NF
 - $BM \rightarrow N$: B, M, N are prime attributes $\rightarrow$ satisfy 3NF
 - $OP \rightarrow B$: B, P, O are prime attributes $\rightarrow$ satisfy 3NF
 - $NO \rightarrow P$: N, O, P are prime attributes $\rightarrow$ satisfy 3NF

So, decomposition $D = \{Q1, Q2\}$ satisfies the 3NF

c) Does D satisfy two properties of lossless join and dependency preservation? Support your answer

$Q(ABDELMNOPX)$ has key AE, LOP, BMLO, and NLO

$F_0 = \{AE \rightarrow LMOPX, LOP \rightarrow AE, BM \rightarrow N, OP \rightarrow BD, NO \rightarrow P\}$

Q1 (OPD) has key OP

$F1 = \{OP \rightarrow D\}$

Q2 (AEBLMNOPX) has key AE, LOP, BMLO, and NLO

$F2 = \{AE \rightarrow LMOPX, LOP \rightarrow AE, BM \rightarrow N, OP \rightarrow B, NO \rightarrow P\}$

- Lossless join

	A	B	D	E	L	M	N	O	P	X
Q1	b11	b12	b13	b14	b15	b16	b17	b18	b19	b110
Q2	b21	b22	b23	b24	b25	b26	b27	b28	b29	b210

	A	B	D	E	L	M	N	O	P	X
Q1	b11	b12	a3	b14	b15	b16	b17	a8	a9	b110
Q2	a1	a2	b23	a4	a5	a6	a7	a8	a9	a10

Use $OP \rightarrow D$

	A	B	D	E	L	M	N	O	P	X
Q1	b11	b12	a3	b14	b15	b16	b17	a8	a9	b110
Q2	a1	a2	a3	a4	a5	a6	a7	a8	a9	a10

Since all of the letters "a" are present in the second row of the table, we may conclude that the decomposition D fulfills the lossless join property.

- Dependency preservation:

The decomposition D satisfies property of dependency preservation as

$F_0 = \{f_1: AE \rightarrow LMOPX, f_2: LOP \rightarrow AE, f_3: BM \rightarrow N, f_4: OP \rightarrow BD, f_5: NO \rightarrow P\}$

Q1 (OPD) has key OP

$F_1 = \{OP \rightarrow D\}$

Q2 (AEBLMNOPX) has key AE, LOP, BMLO, and NLO

$F_2 = \{AE \rightarrow LMOPX, LOP \rightarrow AE, BM \rightarrow N, OP \rightarrow B, NO \rightarrow P\}$

We have $F' = F_1 \cup F_2 = \{OP \rightarrow D, AE \rightarrow LMOPX, LOP \rightarrow AE, BM \rightarrow N, OP \rightarrow B, NO \rightarrow P\} = \{AE \rightarrow LMOPX, LOP \rightarrow AE, BM \rightarrow N, OP \rightarrow BD, NO \rightarrow P\}$

We have $F' \equiv F_0$, so the decomposition D satisfies dependency preservation.